

Blood in Urine

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Blood in the urine (hematuria) can make urine appear pink, red, or brown, depending on the amount of blood, how long it has been in the urine, and how acidic the urine is. An amount of blood too small to change color of the urine (microscopic hematuria) may be found by chemical tests or microscopic examination. Microscopic hematuria may be found when a urine test is done for another reason. (See also [Asymptomatic Proteinuria and Hematuria](#)).

People with hematuria may have other symptoms of [urinary tract disorders](#), such as pain in the side or back (flank), lower abdominal pain, an urgent need to urinate, or difficulty urinating, depending on the cause of blood in the urine. If sufficient blood is present in the urine, the blood may form a clot. The clot can completely block the flow of urine, causing sudden extreme pain and the inability to urinate. Bleeding severe enough to cause such a clot is usually caused by an injury to the urinary tract.

Red urine is not always caused by red blood cells. Red or reddish brown discoloration may also result from the following:

- Hemoglobin (which carries oxygen in red blood cells) in the urine due to the breakdown of red blood cells
- Muscle protein (myoglobin) in urine due to the breakdown of muscle cells
- Porphyria (a disorder caused by deficiencies of enzymes involved in the production of heme, a chemical compound that contains iron and gives blood its red color)
- Foods (for example, beets, rhubarb, and sometimes food coloring)
- Drugs (most commonly [phenazopyridine](#), but sometimes cascara, diphenylhydantoin, [methyldopa](#), [rifampin](#), phenacetin, phenothiazines, and [senna](#))

Causes of Blood in Urine

Blood in the urine may be caused by problems anywhere along the urinary tract from the kidneys to the ureters, bladder, or urethra. Some women at first mistake [vaginal bleeding](#) for blood in the urine.

Common causes

The most common causes differ somewhat by the person's age but overall are

- [Bladder infection](#) (cystitis)
- [Prostate infection](#) (prostatitis)
- [Urinary tract stones](#) (in adults)
- [Kidney and urinary tract birth defects](#)

Less common causes

Less common causes include

- Cancer (of the [kidneys](#), [bladder](#), or [prostate](#))
- Noncancerous enlargement of the prostate ([benign prostatic hyperplasia](#))
- Disorders of the small blood vessels of the kidneys (called [kidney filtering disorders](#) or glomerular disorders)
- Cysts in the kidneys ([polycystic kidney disease](#))
- Narrowing scars (called [strictures](#)) or other abnormalities of the ureters

Cancer and benign prostatic hyperplasia may cause blood in the urine. These disorders are a concern mainly in people over 50, although younger people with risk factors (smoking, family history, or chemical exposures) may develop cancer.

Disorders of the microscopic blood vessels of the kidneys (glomeruli) can be a cause at any age.

[Kidney filtering disorders](#) (glomerular disorders) may be part of a kidney disorder or may occur as a result of a disorder elsewhere in the body. These disorders are more likely if the urine has protein, clumps of red blood cells (called red blood cell casts), or malformed red blood cells. Such disorders include infections (such as a heart valve infection), connective tissue disorders (such as [systemic lupus erythematosus](#)) and [vasculitis](#), blood disorders (such as serum sickness), or certain chronic disorders (such as [diabetes](#)). Also, almost any kind of kidney damage may cause small amounts of blood in the urine.

Severe injuries, such as from a fall or a motor vehicle crash, can injure the kidneys or bladder and cause bleeding. Various procedures and surgeries (for example, inserting a catheter or doing a prostate or kidney biopsy) can also cause bleeding.

Schistosoma haematobium is a parasitic worm that causes a disease called [schistosomiasis](#) in Africa and, to a lesser extent, in India and parts of the Middle East. This worm can invade the urinary tract, causing blood in the urine. Doctors consider schistosomiasis only if people have spent time in areas where the worm is found. [Tuberculosis](#) may cause blood in the urine.

Other causes of blood in the urine include bladder inflammation from radiation therapy of the pelvic region, certain drugs, and [blood vessel disorders of the kidney](#).

Because blood in the urine of a person taking an anticoagulant (a "blood-thinner" that prevents blood clots) is not necessarily caused by that medication, the doctor's evaluation will take all other potential causes into consideration.

Evaluation of Blood in Urine

Doctors first try to establish that bleeding is the cause of red urine. Then they look for the cause of the bleeding, including where in the urinary tract (or occasionally elsewhere) the bleeding is originating. The following information can help people know when to see a doctor and what to expect during the evaluation.

Warning signs

In people with blood in the urine, certain symptoms and characteristics are cause for concern. They include

- Large amount of blood in the urine
- Age over 50
- Swelling of the feet or legs, plus [high blood pressure](#)

When to see a doctor

People who notice blood in their urine should see their doctor within 1 or 2 days. However, people who are passing a large amount of blood, who are unable to urinate, or who have severe pain should see a doctor right away.

What the doctor does

Doctors first ask questions about the person's symptoms and medical history and then do a physical examination. What they find during the history and physical examination often suggests a cause of the blood in the urine and the tests that may need to be done (see table [Some Causes and Features of Blood in the Urine](#))

Doctors ask how long blood has been present and whether there have been any previous bleeding episodes. They ask about fever, weight loss, or symptoms of urinary blockage, such as difficulty starting urination or inability to completely empty the bladder. Pain or discomfort is an important finding. Burning during urination or dull pain in the lower abdomen just above the pubic bone suggests a bladder infection. In men, mild to moderate pain in the lower back or pelvis is often the result of a [prostate infection](#). Extremely severe pain is usually due to a stone or a blood clot blocking the flow of urine.

Doctors then do a physical examination. Usually, a pelvic examination is necessary in women. If women have blood in the vagina, a catheter may need to be inserted into the bladder to see whether the source of blood is the bladder or the vagina. In men, doctors usually do a digital rectal examination to check the prostate.



Some Causes and Features of Blood in the Urine

Cause	Common Features*	Tests†
Benign prostatic hyperplasia (noncancerous enlargement of the prostate gland)	Mainly in men over 50 Often difficulty starting urination, a weak urine stream, a sensation of incomplete urination, or dribbling at the end of urination An enlarged prostate detected during a digital rectal examination	Blood tests to measure the PSA level Cystoscopy Often ultrasonography of the bladder to measure how much urine remains in the bladder after voiding (postvoid residual urine volume)
Bladder cancer or kidney cancer	Mainly in people over 50 or with risk factors for these cancers (smoking, family members who have had cancer, or exposure to chemicals that may cause cancer) Sometimes burning or pain during urination or an urgent need to urinate	Examination of the interior of the bladder using a flexible viewing tube inserted through the urethra (cystoscopy) Possible bladder biopsy Sometimes CT or MRI
	Often symptoms that affect the whole body (such as fever, chills, weight loss or sweating)	If prostate cancer is suspected, prostate biopsy
Cystitis (bladder infection)	Usually in women and girls	A doctor's examination



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